```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on e.ssn=s.ssn group by s.ssn, s.firstNam
e, s.lastName having count(e.courseId)=5;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Kevin     | Green    | 100000041 |
| Michelle  | Adams    | 100000042 |
| Brian     | Nelson   | 100000043 |
| Carol     | Baker    | 100000044 |
| Edward    | Hall     | 100000045 |
| Ronald    | Campbell | 100000047 |
| Melissa   | Mitchell | 100000048 |
| Timothy   | Carter   | 100000049 |
| Deborah   | Roberts  | 100000050 |
| Stephanie | Phillips | 100000052 |
| Jeffrey   | Evans    | 100000053 |
| Rebecca   | Turner   | 100000054 |
| Ryan      | Diaz     | 100000055 |
| Laura     | Parker   | 100000056 |
| Sharon    | Edwards  | 100000058 |
| Gary      | Collins  | 100000059 |
| Cynthia   | Reyes    | 100000060 |
+-----------+----------+-----------+
17 rows in set (0.006 sec)
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on e.ssn=s.ssn group by s.ssn, s.firstNam
e, s.lastName having count(e.courseId)=4;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Amanda    | Rivera   | 100000046 |
| Jacob     | Cruz     | 100000057 |
+-----------+----------+-----------+
2 rows in set (0.001 sec)
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on e.ssn=s.ssn group by s.ssn, s.firstNa
e, s.lastName having count(e.courseId)=3;
+-------------+-----------+-----------+
| firstName   | lastName  | ssn       |
+-------------+-----------+-----------+
| James       | Smith     | 100000001 |
| Mary        | Johnson   | 100000002 |
| Robert      | Williams  | 100000003 |
| Patricia    | Brown     | 100000004 |
| John        | Jones     | 100000005 |
| Jennifer    | Garcia    | 100000006 |
| Michael     | Miller    | 100000007 |
| Linda       | Davis     | 100000008 |
| David       | Rodriguez | 100000009 |
| Elizabeth   | Martinez  | 100000010 |
| William     | Hernandez | 100000011 |
| Barbara     | Lopez     | 100000012 |
| Richard     | Gonzalez  | 100000013 |
| Susan       | Wilson    | 100000014 |
| Joseph      | Anderson  | 100000015 |
| Jessica     | Thomas    | 100000016 |
| Thomas      | Taylor    | 100000017 |
| Sarah       | Moore     | 100000018 |
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on e.ssn=s.ssn group by s.ssn, s.firstNam
e, s.lastName having count(e.courseId)=2;
+-----------+------------+-----------+
| firstName | lastName   | ssn       |
+-----------+------------+-----------+
| Nicholas  | Stewart    | 100000061 |
| Kathleen  | Morris     | 100000062 |
| Eric      | Morales    | 100000063 |
| Amy       | Murphy     | 100000064 |
| Stephen   | Cook       | 100000065 |
| Shirley   | Rogers     | 100000066 |
| Jonathan  | Gutierrez  | 100000067 |
| Angela    | Ortiz      | 100000068 |
| Larry     | Morgan     | 100000069 |
| Helen     | Cooper     | 100000070 |
| Justin    | Peterson   | 100000071 |
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on e.ssn=s.ssn group by s.ssn, s.firstNam
e, s.lastName having count(e.courseId)=1;
+-----------+----------+-----------+
| firstName | lastName | ssn       |
+-----------+----------+-----------+
| Charles   | Lee      | 100000021 |
| Nancy     | Perez    | 100000022 |
| Matthew   | Thompson | 100000023 |
| Lisa      | White    | 100000024 |
| Anthony   | Harris   | 100000025 |
| Betty     | Sanchez  | 100000026 |
| Mark      | Clark    | 100000027 |
| Margaret  | Ramirez  | 100000028 |
| Donald    | Lewis    | 100000029 |
| Sandra    | Robinson | 100000030 |
| Steven    | Walker   | 100000031 |
| Ashley    | Young    | 100000032 |
| Paul      | Allen    | 100000033 |
| Dorothy   | King     | 100000034 |
| Andrew    | Wright   | 100000035 |
| Kimberly  | Scott    | 100000036 |
| Joshua    | Torres   | 100000037 |
| Emily     | Nguyen   | 100000038 |
| Kenneth   | Hill     | 100000039 |
| Donna     | Flores   | 100000040 |
+-----------+----------+-----------+
20 rows in set (0.001 sec)
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn from StudentTable s join EnrollmentTable e on e.ssn=s.ssn group by s.ssn, s.firstNam
e, s.lastName having count(e.courseId)=0;
Empty set (0.003 sec)
```

```
MariaDB [mynewpaltz]> select c.subjectId, count(*) as count from CourseTable c where c.numOfCredits = 4 group by c.subjectId order by c.subjectId;
+-----------+-------+
| subjectId | count |
+-----------+-------+
| CS        |     2 |
| ENGR      |     2 |
| MATH      |     2 |
+-----------+-------+
3 rows in set (0.001 sec)

MariaDB [mynewpaltz]> select c.subjectId, count(*) as count from CourseTable c where c.numOfCredits = 3 group by c.subjectId order by c.subjectId;
+-----------+-------+
| subjectId | count |
+-----------+-------+
| CS        |     3 |
| ENGL      |     5 |
| ENGR      |     2 |
| MATH      |     3 |
+-----------+-------+
4 rows in set (0.001 sec)

MariaDB [mynewpaltz]> select c.subjectId, count(*) as count from CourseTable c where c.numOfCredits = 2 group by c.subjectId order by c.subjectId;
+-----------+-------+
| subjectId | count |
+-----------+-------+
| ENGR      |     1 |
+-----------+-------+
1 row in set (0.001 sec)

MariaDB [mynewpaltz]> select c.subjectId, count(*) as count from CourseTable c where c.numOfCredits = 1 group by c.subjectId order by c.subjectId;
Empty set (0.001 sec)

MariaDB [mynewpaltz]> select c.subjectId, count(*) as count from CourseTable c where c.numOfCredits = 5 group by c.subjectId order by c.subjectId;
Empty set (0.001 sec)
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, s.ssn, c.courseId, c.title, e.grade from EnrollmentTable e join St
udentTable s on s.ssn=e.ssn join CourseTable c on c.courseId=e.courseId order by s.firstName, s.lastName, c.courseNumber
, c.title;
```

firstName	lastName	ssn	courseId	title	grade
Aaron	West	100000099	3	Linear Algebra	B
Aaron	West	100000099	8	Operating Systems	B
Alexander	Diaz	100000089	18	Dynamics	A
Alexander	Diaz	100000089	13	World Literature	C
Amanda	Rivera	100000046	6	Intro to Computer Science	B
Amanda	Rivera	100000046	2	Calculus II	A
Amanda	Rivera	100000046	4	Probability and Statistics	A
Amanda	Rivera	100000046	8	Operating Systems	B
Amy	Murphy	100000064	13	World Literature	B
Amy	Murphy	100000064	8	Operating Systems	A
Andrew	Wright	100000035	15	Shakespeare	C
Angela	Ortiz	100000068	12	English Composition II	C
Angela	Ortiz	100000068	17	Statics	A
Anna	Bailey	100000072	16	Intro to Engineering	B
Anna	Bailey	100000072	1	Calculus I	B
Anthony	Harris	100000025	5	Abstract Algebra	B
Ashley	Young	100000032	12	English Composition II	C
Barbara	Lopez	100000012	2	Calculus II	A
Barbara	Lopez	100000012	12	English Composition II	C
Barbara	Lopez	100000012	17	Statics	B
Benjamin	Alvarado	100000079	3	Linear Algebra	A
Benjamin	Alvarado	100000079	8	Operating Systems	C

```
MariaDB [mynewpaltz]> select c.courseId, c.courseNumber, c.subjectId, count(e.ssn) from CourseTable c left join Enrollme
ntTable e on e.courseId=c.courseId where e.courseId=10;
+----------+--------------+-----------+--------------+
| courseId | courseNumber | subjectId | count(e.ssn) |
+----------+--------------+-----------+--------------+
|       10 |          450 | CS        |           13 |
+----------+--------------+-----------+--------------+
1 row in set (0.000 sec)
```

```
MariaDB [mynewpaltz]> select c.courseId, c.courseNumber, c.subjectId, c.title, count(e.ssn) as students from CourseTable
 c left join EnrollmentTable e on e.courseId=c.courseId group by c.courseId;
+----------+--------------+-----------+----------------------------+----------+
| courseId | courseNumber | subjectId | title                      | students |
+----------+--------------+-----------+----------------------------+----------+
|        1 |          101 | MATH      | Calculus I                 |       13 |
|        2 |          102 | MATH      | Calculus II                |       13 |
|        3 |          201 | MATH      | Linear Algebra             |       12 |
|        4 |          300 | MATH      | Probability and Statistics |       14 |
|        5 |          405 | MATH      | Abstract Algebra           |       13 |
|        6 |          101 | CS        | Intro to Computer Science  |       11 |
|        7 |          210 | CS        | Data Structures            |       13 |
|        8 |          315 | CS        | Operating Systems          |       13 |
|        9 |          402 | CS        | Database Management Systems |       12 |
|       10 |          450 | CS        | Artificial Intelligence    |       13 |
|       11 |          101 | ENGL      | English Composition I      |       14 |
|       12 |          102 | ENGL      | English Composition II     |       13 |
|       13 |          205 | ENGL      | World Literature           |       14 |
|       14 |          310 | ENGL      | Creative Writing           |       14 |
|       15 |          415 | ENGL      | Shakespeare                |       12 |
|       16 |          100 | ENGR      | Intro to Engineering       |       12 |
|       17 |          201 | ENGR      | Statics                    |       12 |
|       18 |          202 | ENGR      | Dynamics                   |       12 |
|       19 |          305 | ENGR      | Thermodynamics             |       11 |
|       20 |          410 | ENGR      | Circuit Analysis           |       10 |
+----------+--------------+-----------+----------------------------+----------+
20 rows in set (0.001 sec)
```

```
MariaDB [mynewpaltz]> select s.firstName, s.lastName, sum(c.numOfCredits) from EnrollmentTable e join StudentTable s on
s.ssn=e.ssn join CourseTable c on c.courseId=e.courseId group by s.firstName, s.lastName order by s.firstName, s.lastNam
e, count(c.numOfCredits);
+-------------+------------+--------------------+
| firstName   | lastName   | sum(c.numOfCredits) |
+-------------+------------+--------------------+
| Aaron       | West       |                  7 |
| Alexander   | Diaz       |                  6 |
| Amanda      | Rivera     |                 15 |
| Amy         | Murphy     |                  7 |
| Andrew      | Wright     |                  3 |
| Angela      | Ortiz      |                  6 |
| Anna        | Bailey     |                  6 |
| Anthony     | Harris     |                  3 |
```